AI Agent — Enterprise Knowledge Assistant

Team Focus: 1 (backend architect) **Primary Language:** C# / Python **Key Framework:** Microsoft Agent Framework / Semantic Kernel

Business Scenario

Your company has accumulated years of project documentation, technical proposals, contracts, and email threads. Employees waste 15+ minutes per day searching for historical information.

Goal: Build an AI Agent that understands natural language queries, autonomously retrieves relevant documents, and generates accurate answers with citations.

Example query: "What was the pricing model we offered to Client X last year?"

Technology Stack

Layer	.NET Path	Python / Alt Path
Agent Framework	Microsoft Agent Framework (C#)	Microsoft Agent Framework / LangChain
LLM	Azure OpenAI / OpenAI SDK	Azure OpenAI / OpenAI
Vector DB	Azure AI Search	Azure AI Search / Pinecone
Orchestration	ASP.NET Core Web API	FastAPI

Microsoft Agent Framework combines AutoGen's simple agent abstraction with Semantic Kernel's enterprise features — session-based state management, type safety, middleware, and telemetry — and integrates directly with ASP.NET Core dependency injection.

Learning Path (4 Weeks)

Week	Task	Tech Stack
1	Set up .NET 10 SDK Web API project; call Azure OpenAI for basic Q&A	.NET 10, Azure OpenAI SDK
2	Implement RAG: chunk PDFs, vectorize, store in Azure AI Search	Azure AI Search, Semantic Kernel
3	Add tool calling: agent can query internal databases and APIs autonomously	Microsoft Agent Framework function calling
4	Add conversation memory, user feedback loop, deploy to production	Redis Cache, Application Insights

Suggested Enhancements — Added Business Value

These go beyond the base build and are what turn a good prototype into a compelling, differentiated product:

- **Source-level citations:** Return clickable links to the exact paragraph/page in the original document, not just the file name — builds user trust.
- **Multi-format ingestion:** Extend beyond PDFs to Word docs, Excel sheets, Slack/Teams exports, and scanned images (OCR) so the assistant covers all real company knowledge.
- **Feedback-driven re-ranking:** Thumbs up/down on answers feeds a lightweight re-ranking model so answer quality improves over time.
- **Access-control aware retrieval:** Respect document-level permissions so the agent never surfaces content a given employee shouldn't see.
- **Teams / Slack integration:** Let employees ask questions from where they already work instead of a separate portal.
- **Usage analytics dashboard:** Track most-asked questions, knowledge gaps, and time saved — turns the tool into a measurable ROI story.

Key Resources

- Microsoft Agent Framework Quick Start — learn.microsoft.com/en-us/agent-framework/tutorials/quick-start
- Agents in Workflows Tutorial — learn.microsoft.com/en-us/agent-framework/tutorials/workflows/agents-in-workflows
- Semantic Kernel RAG Implementation (GitHub: vibbs/agent-orchestration-fw-rag)
- MAF Getting Started Samples (GitHub: NikiforovAll/maf-getting-started)

Each project should still produce: a working prototype, a blog post, a GitHub repository, a recorded demo video, and an architecture diagram.